FINAL PROJECT ASSIGNMENT – PART 2

ISDS 7075

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PREDICTING WORKOUT BURNOUT AND OVERTRAINING RISK USING FITBIT ACTIVITY AND RECOVERY DATA

EXECUTIVE SUMMARY

The objective of this project is to develop a machine learning model capable of identifying patterns associated with workout burnout and sustained activity decline using Fitbit activity, sleep, and recovery data. During Part 1, the Fitbit datasets were explored, cleaned, merged, and prepared for machine learning. In Part 2, multiple classification models were developed and compared to determine which approach most effectively identified sustained reductions in activity.

Three machine learning models: Logistic Regression, Decision Tree, and Random Forest, were evaluated using accuracy, precision, recall, and F1-score. Although Random Forest achieved the highest overall accuracy, the Decision Tree produced the highest F1-score and provided the best balance between balancing precision and recall when identifying sustained activity decline. Hyperparameter tuning was then performed using GridSearchCV to determine whether additional improvements could be achieved. While tuning produced similar performance, the original Decision Tree remained the strongest model for this dataset.

INTRODUCTION

Many people begin an exercise program with high motivation and activity levels. Over time, maintaining demanding workout routines without sufficient recovery may contribute to fatigue, declining consistency, and eventual disengagement. Wearable devices continuously collect information about daily movement and sleep that may reveal early behavioral changes before users significantly reduce their activity.

This project explores whether Fitbit-generated activity, intensity, and sleep measures can be organized into a reliable analytical dataset and used to identify sustained reductions in exercise behavior. Part 1 focused on problem framing, exploratory analysis, and data preparation. Part 2 builds upon that work by developing, comparing, and tuning multiple machine learning models to determine which approach best predicts sustained activity decline.

Building on the exploratory analysis completed in Part 1, this phase focuses on selecting appropriate machine learning models, comparing their predictive performance, tuning the most promising model, and evaluating the results using standard classification metrics.

PROBLEM STATEMENT

The objective of this project is to develop a machine learning model that predicts periods of workout burnout or sustained activity decline using Fitbit-generated activity, sleep, and intensity data. During Part 1, the datasets were explored, cleaned, merged, and prepared for analysis. Building on that foundation, Part 2 focuses on identifying the machine learning model that best predicts sustained activity decline by comparing multiple classification approaches and optimizing the selected model through hyperparameter tuning. Our team evaluated Logistic Regression, Decision Tree, and Random Forest models using standard classification metrics, including accuracy, precision, recall, and F1-score, to determine which model provides the best overall performance. The analysis utilizes the Daily_activity_clean.csv, Sleep_clean.csv, and hourly_intensities_clean.csv datasets to support model development and evaluation.

BUSINESS VALUE: WHY THIS PROBLEM MATTERS

Fitbit devices continuously collect activity, sleep, and movement data that many users never fully utilize beyond daily step counts or calories burned. Applying machine learning to these data provides an opportunity to identify behavioral patterns that may indicate declining exercise consistency before users completely disengage from their fitness routines.

Early identification of sustained activity decline could help users better balance exercise and recovery, potentially reducing overtraining while encouraging long-term adherence to healthy behaviors. Although this project does not directly measure burnout, it demonstrates how wearable-device data can be leveraged to identify behavioral patterns associated with declining activity.

WHAT SUCCESS LOOKS LIKE

A successful outcome for this project is identifying a machine learning model that can reasonably distinguish between participants maintaining consistent activity levels and those experiencing sustained activity decline. Success is measured by comparing multiple classification models using Accuracy, Precision, Recall, and F1-score, then selecting the model that provides the best overall balance of performance. An additional measure of success is determining whether hyperparameter tuning improves the selected model while maintaining consistent performance on previously unseen test data.

ASSUMPTIONS:

Several assumptions guide this project and will be evaluated as additional modeling is completed.

ASSUMPTION	VERIFICATION
Higher activity combined with poor sleep increases burnout risk	Analyzing workout duration, calories burned, and active minutes with sleep duration and quality.
Users who are approaching burnout show declining activity before quitting	Analyze activity trends before prolonged inactivity
Fitbit metrics capture recovery behavior	Explore relationships among sleep, calories, intensity, and activity
Participant ID and date provide reliable keys for combining datasets.	Validate merges after standardizing all date fields and confirming participant-day alignment.

DATA SOURCES

The analysis utilized three publicly available Fitbit datasets obtained from Kaggle:

- Daily Activity
- Hourly Intensities
- Sleep Data

These datasets were merged using participant ID and date to create a single analytical dataset used throughout the modeling process.

DATA PREPARATION

The datasets required only minor preprocessing before modeling. Data cleaning steps included:

- Removing unnecessary variables such as the **rn** column.
- Converting Activity_Hour and Sleep_Day into datetime objects.
- Checking for duplicate observations.
- Aggregating hourly intensity data into daily participant summaries.
- Merging activity, sleep, and intensity datasets.
- Creating engineered features including prior activity averages, prior sedentary change, activity-distance tiers, and the sustained activity decline target variable.

The final dataset contained activity, sleep, movement, and engineered behavioral variables suitable for machine learning analysis.

EXPLORATORY DATA ANALYSIS

Exploratory analysis confirmed that the Fitbit data were generally clean and well structured for modeling. Correlation analysis identified several highly related movement variables, supporting the removal of redundant predictors before model development. Variable distributions demonstrated substantial variability across participants while remaining appropriate for machine

learning analysis. These findings supported the selection of activity and recovery variables used to predict sustained activity decline.

MODEL COMPARISON

Three classification models were evaluated: Logistic Regression, Decision Tree, and Random Forest. Performance was compared using Accuracy, Precision, Recall, and F1-score. While Random Forest achieved the highest overall accuracy (0.915), its lower recall for sustained activity decline resulted in a lower F1-score than the Decision Tree. Logistic Regression demonstrated similar overall performance but produced more false positive classifications. Because identifying sustained activity decline requires balancing both precision and recall, the Decision Tree produced the strongest overall performance with an F1-score of 0.592 and was selected for further tuning. Accuracy alone was not used to select the final model because sustained activity decline represented the minority class in the dataset.

	Model	Accuracy	Precision	Recall	F1 Score
0	Decision Tree	0.883	0.512	0.700	0.592
1	Logistic Regression	0.855	0.440	0.733	0.550
2	Random Forest	0.915	0.800	0.400	0.533

MODEL TUNING

The selected Decision Tree model was further optimized using GridSearchCV to evaluate multiple combinations of maximum tree depth, minimum samples required for node splits, minimum samples per leaf, and split criteria. Hyperparameter tuning was evaluated using five-fold cross-validation with the F1-score selected as the primary evaluation metric due to the relatively small number of sustained decline observations.

Although the tuned Decision Tree produced similar overall performance, it did not improve upon the original model. This demonstrates that tuning does not always improve performance and reinforces the importance of evaluating tuned models on an independent testing dataset. The original Decision Tree maintained the highest F1-score and therefore remained the final model selected for this project.

	Model	Accuracy	Precision	Recall	F1 Score
0	Original Decision Tree	0.883	0.512	0.7	0.592
1	Tuned Decision Tree	0.875	0.488	0.7	0.575

TARGET VARIABLE

Because the Fitbit datasets do not contain a direct measure of workout burnout, a behavioral target variable was created to represent sustained activity decline. Rather than predicting clinical burnout, the model identifies periods where participants exhibit meaningful reductions in physical activity over time.

This target serves as a proxy for potential workout burnout and allows machine learning models to identify behavioral patterns associated with declining exercise consistency. While additional validation is still needed, this approach provides a practical method for evaluating workout burnout risk using the available Fitbit data.

LIMITATIONS

Several limitations should be considered when interpreting the results of this analysis. Most importantly, the available Fitbit data do not contain a direct measure of workout burnout, overtraining, or participant motivation. Consequently, the models developed during this phase predict sustained activity decline rather than a clinically validated measure of workout burnout.

Sleep information is available for only a subset of participants, and the observation period covers approximately one month. Additionally, repeated observations from the same participants may introduce individual behavioral patterns that are not representative of the broader population. These limitations reinforce that the current analyses are exploratory and primarily intended to prepare the data for future predictive modeling.

CONCLUSION

This phase of the project focused on developing and evaluating machine learning models capable of identifying sustained activity decline using Fitbit activity and recovery data. After comparing Logistic Regression, Decision Tree, and Random Forest models, the Decision Tree was selected as the strongest overall model because it achieved the highest F1-score while maintaining a good balance between precision and recall. Hyperparameter tuning was performed using GridSearchCV to evaluate whether additional improvements could be achieved; however, the original Decision Tree remained the best-performing model for this dataset.

The results from this analysis demonstrate that Fitbit activity and recovery metrics contain meaningful patterns that can be used to identify sustained declines in physical activity. While additional validation and refinement are still needed, the selected model provides a strong foundation for continuing the analysis and further evaluating its ability to identify potential workout burnout risk.

DATA VISUALIZATIONS

FIGURE 1. Correlation Matrix of Fitbit Activity Metrics

Strong positive correlations were observed between several movement variables, particularly Total Distance and Tracker Distance. These relationships indicated redundant information, supporting feature selection before model development.

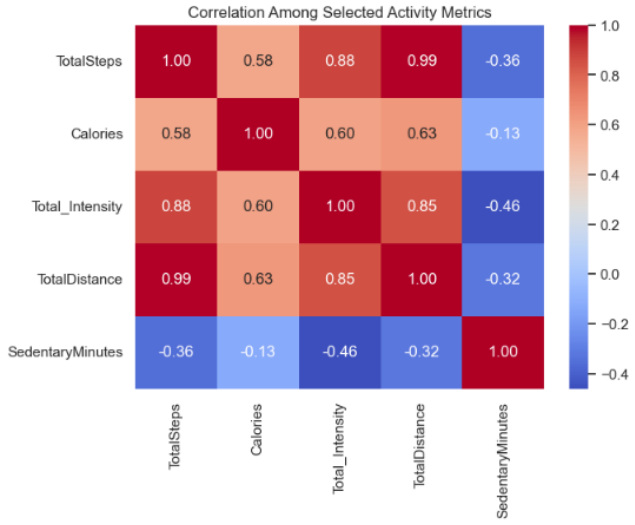


FIGURE 2. Distribution of Key Fitbit Activity and Recovery Metrics

The distributions of daily steps, calories, activity intensity, and sleep duration demonstrate considerable variability across participants while showing no major abnormalities that would prevent machine learning analysis.

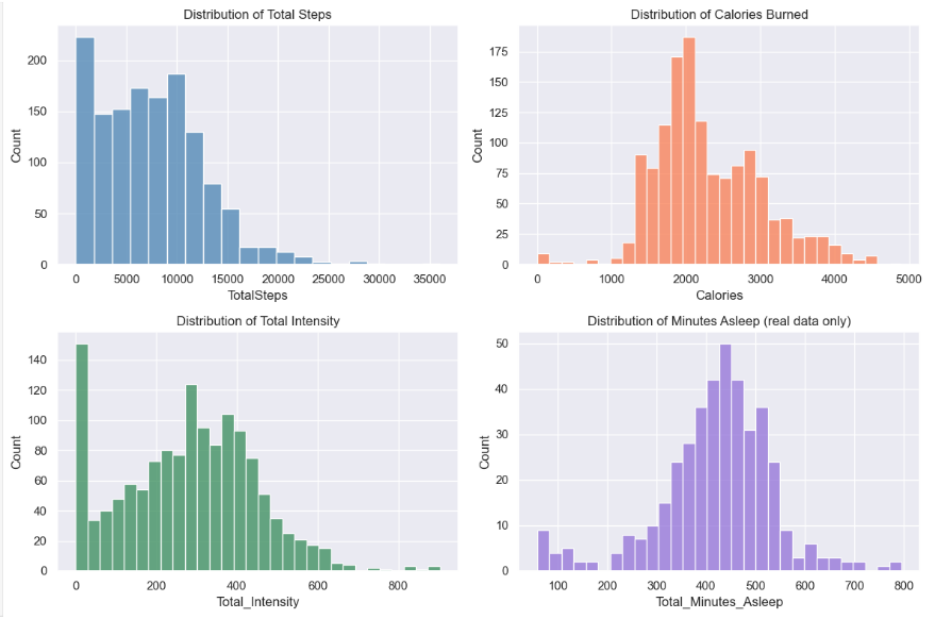


FIGURE 3. CONFUSION MATRICES FOR CANDIDATE CLASSIFICATION MODELS

Figure 3 compares the performance of the three candidate classification models: Logistic Regression, Decision Tree, and Random Forest. While Random Forest achieved the highest overall accuracy by correctly identifying most "No Decline" observations, it detected fewer sustained decline cases (12 of 30). Logistic Regression identified more decline cases but produced more false positives. The Decision Tree provided the strongest balance between correctly identifying both classes, which resulted in the highest F1-score (0.592). Because the primary goal of this project is to identify sustained activity decline rather than simply maximize overall accuracy, the Decision Tree was selected for further tuning.

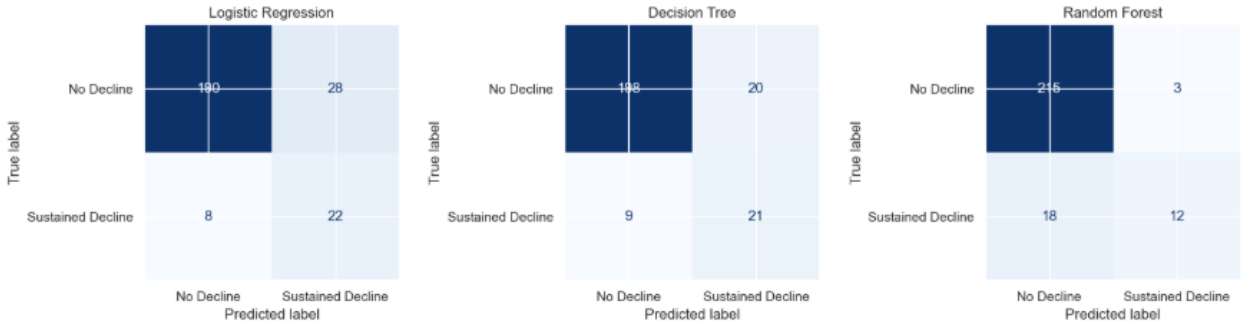


FIGURE 4. Confusion Matrix for the Final Decision Tree Model

The final Decision Tree correctly classified 198 observations with no sustained decline and 21 observations with sustained decline. Nine decline cases were missed (false negatives), while 20 observations were incorrectly classified as sustained decline (false positives). Overall, the model achieved an accuracy of 88.3% and the highest F1-score among the candidate models, making it the best overall choice for this project.

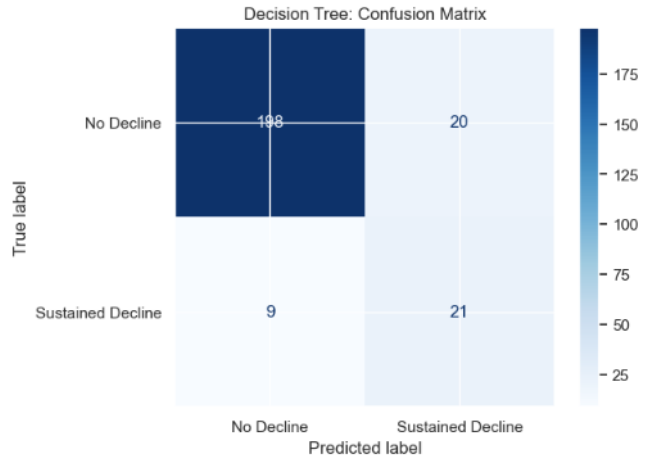
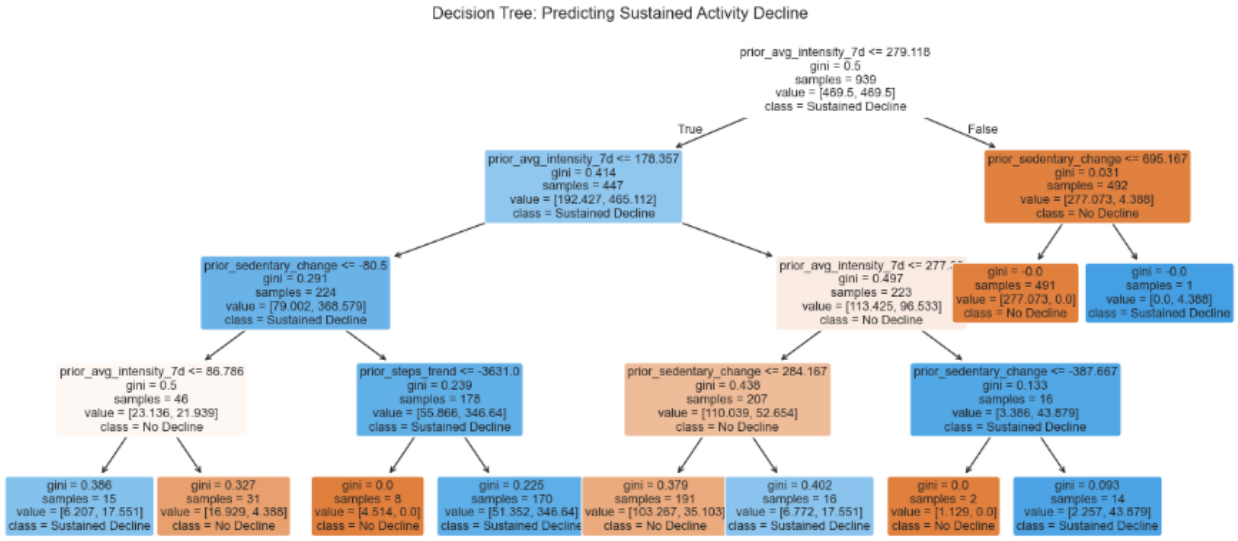


FIGURE 5. Final Decision Tree Model

The final Decision Tree uses prior average intensity, changes in sedentary behavior, and previous step trends to classify whether a participant is likely to experience sustained activity decline. Variables appearing near the top of the tree have the greatest influence on classification decisions. The tree also demonstrates that multiple activity and recovery measures work together rather than relying on a single metric to make predictions.



REFERENCES

Elohorigbine. *Bellabeat Cleaned Analysis Data*. Kaggle, https://www.kaggle.com/datasets/elohorigbine/bellabeat-cleaned-analysis-data/data. Accessed 29 July 2026.